

Dr. Vipul Silwal

Mailing Address

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Education

PhD in Geophysics 2012-2018
University of Alaska Fairbanks, USA
Thesis: Earthquake source mechanisms and three-dimensional wavefield simulations in Alaska
Advisor: Dr. Carl Tape

Integrated BS and MS, Exploration Geophysics 2007-2012
Indian Institute of Technology (IIT), Kharagpur, India
Thesis: Moment Tensor Inversion in Alaska using Body and Surface Waves
Advisor: Dr. William Mohanty and Dr. Carl Tape

Experience

Assistant Professor

Department of Earth Sciences,
Indian Institute of Technology Roorkee, India June 2019 - present

Joint Faculty

International Center of Excellence for Dams,
Indian Institute of Technology Roorkee, India June 2023 - present

Research Associate

Department of Civil Engineering
Indian Institute of Technology Hyderabad, India Mar 2019 - May 2019

Research Intern

University of Tromso, Norway Summer 2010

Teaching

IITR - PhD Students

1. Rinku Mahanta - Waveform-based Moment tensor inversion in Himalayas (ongoing)
2. Pankaj Lahon- 3D wavefield simulations and seismic imaging in Himalayas (ongoing)
3. Nivika Singh - Waveform-based source location of icequakes (ongoing)

Courses - IITR

1. ESN-421 Numerical modeling in Geosciences (Autumn)
2. ESN-510 Numerical techniques and computer programming (Autumn)
3. ESN-102 Global Geophysics (Autumn)
4. ESN-422 Geophysical Inversion (Spring)
5. ESN-472 Advanced Seismology (Spring)
6. DSR-505 Dam Safety Surveillance, Instrumentation and Monitoring (Spring)

Funded Project (Research/ Consultancy)

1. Seismic ambient noise and P-wave coda correlation imaging in the Bikaner-Nagaur basin, Rajasthan for detailed crustal structure (co-PI) 2026-2028
2. Seismic Vibration Criteria Evaluation for Site Assessment, ReNew Photovoltaic Pvt Ltd. 2025
3. Moment tensor Inversion using 3D Green's Functions: Investigating the effects of Crustal thickening in Himalayas on Earthquake Mechanisms, MoES 2025-2026
4. Virtual Lab on Fundamentals of Seismology, Ministry of Education 2025-2026
5. Ultrasonic Seismic imaging application to Sandalwood trees, DS Ltd. 2023-2024
6. Seismic imaging of central Himalayas using spectral element methodology, SERB-SRG 2021-2024
7. Moment tensor inversion and three-dimensional wavefield simulation in Himalayas, IIT-FIG 2020-2023
8. Geological and Geophysical investigation of Bageshwar Mines, Geocoin Ltd. (co-PI) 2022
9. Development of low-cost geophone sensors, iHUB-IITR 2022
10. Bihar 1934 and Uttarkashi 1991 Earthquake scenario Multi-State risk preparedness campaign, NDMA (co-PI) 2020

Publications

1. Vaibhav, S., **Silwal, V***, Yadav, R., (In Press) Seismic-XR: Simulating and Visualizing Seismic Ground Motions in Mixed Reality, Seismological Research Letters.
2. Singh, SP, Padhy, S, and **V Silwal**, (2025), A high-resolution 3D shear velocity model for Cameroon using ambient noise tomography: Constraints from the CPSO algorithm, Geophysical Journal International, <https://doi.org/10.1093/gji/ggaf227>.
3. Mahanta, R., **V Silwal***, Dwivedi, A., Maurya, V. K., and Rajendran, C. P. (2025). Relocation of major earthquakes in India from 1905 to 2011 using a non-linear probabilistic approach. Journal of Asian Earth Sciences, 106632. <https://doi.org/10.1016/j.jseaes.2025.106632>
4. Mahanta, R., **V Silwal***, Sharma, M.L. (2024). Body Waves– and Surface Waves–Derived Moment Tensor Catalog for Garhwal-Kumaon Himalayas. In: Kumar, R., Singh, R., Kanhaiya, S., Maurya, S.P. (eds) Recent Developments in Earthquake Seismology. Springer, Cham. <https://doi.org/10.1007/978-3-031-47538-2-4>
5. Singh, SP and **V Silwal***, (2023), Enhanced crustal and intermediate seismicity in the Hindu Kush-Pamir region revealed by attentive deep learning model, Artificial Intelligence in Geosciences, Volume 4, Pages 150-163.
6. Kumar P, **Silwal V***, Mahanta R, Maurya VK, Kamal, Sharma ML, Ammani A, (2023). Near Real-Time Detection and Moment Tensor Inversion of the 11 May 2022, Dharchula Earthquake. GeoHazards.vol 4(4):515-525. <https://doi.org/10.3390/geohazards4040029>.
7. Kumar V, **V Silwal***, S Ray, (2023), Simulation of Seismic Wave Propagation Using Spectral-Element Method for Uttarakhand Himalayan Region with SPEC-FEM3D. 2nd EAGE/Aqua Foundation Indian Near Surface Geophysics Conference and Exhibition.
8. **V. Silwal***, C. Tape, and A. Lomax, (2018), Crustal earthquakes in the Cook Inlet and Susitna regions, southern Alaska, Tectonophysics, Vol. 745, doi: 10.1016/j.tecto.2018.08.013.
9. Alvizuri, C., **V. Silwal**, L. Krischer, and C. Tape, (2018), Estimation of full moment tensors including uncertainties for earthquakes, volcanic events and nuclear explosions, Journal of Geophysical Research: Solid Earth., Vol. 123, doi: 10.1029/2017JB015325.
10. Tape, C., S. Holtkamp, **V. Silwal**, Y. Kaneko, J. Hawthorne, J. P. Ampuero, N. Ruppert, K. Smith, and M. E. West, (2018), Earthquake nucleation and fault slip complexity in the lower crust of central Alaska, Nature Geoscience, 11, doi: 10.1038/s41561-018-0144-2.

11. Tape, C., A. Lomax, **V. Silwal**, J. D. Agnew and B. Brettschneider, (2017), The 1904 Ms 7.3 Earthquake in Central Alaska, *Bulletin of the Seismological Society of America*, Vol. 107, No. 3, pp. 1147–1174, (2017), doi: 10.1785/0120160178.
12. **V. Silwal*** and C. Tape, (2016), Seismic moment tensors and estimated uncertainties in southern Alaska, *Journal of Geophysical Research: Solid Earth*, v. 121, doi: 10.1002/2015JB012588.
13. C. Tape, **V. Silwal**, C. Ji, L. Keyson, M.E. West, and N. Ruppert, (2015), Transtensional tectonics of the Minto Flats fault zone and Nenana basin, central Alaska, *Bulletin of the Seismological Society of America*, Vol. 105, No. 4, pp. 2081–2100, (2015), doi: 10.1785/0120150055.
14. C. Tape, M. West, **V. Silwal**, and N. Ruppert, (2013), Earthquake nucleation and triggering on an optimally oriented fault, *Earth and Planetary Science Letters*, v. 363, p. 231–241. doi: 10.1016/j.epsl.2012.11.060.
15. D. Trivedi, K. Devi, I. Buynevich, P. Srinivasan, K. Ravisankar, **V. Silwal**, D. Sengupta, and R. R. Nair, (2012), Interpretation of Dune Genesis from the Sedimentological Data and Ground Penetrating Radar (GPR) Signatures: A case study from Ashirmata Dune Field, Mandvi Beach, Gujarat, India, *International Journal of Geosciences*, 2012, 3, 772–779.

(*corresponding author)

Publications- In review

1. Nasara, N., **Silwal V***, Singh, N., (minor revision) RSDB - A Global Seismic Waveform Dataset from the Raspberry Shake Network for Machine Learning Applications, DOI dataset: <https://doi.org/10.5281/zenodo.17265621>.
2. Mahanta, R, **Silwal, V***, Sharma, ML, and Mittal, H, (In review) Moment tensor inversion and uncertainty analysis of earthquakes in Uttarakhand Himalayas, GJI-24-0368.
3. **Silwal, V***, Mahanta, R, Dwivedi, A, and H Mittal, (In review) Nov 3, 2023 Nepal earthquake: Source mechanisms and structural response of Indo-Gangetic basin, *Seismica*.

Reports/ Catalogs

1. Mahanta, R., and **V. Silwal**, (2024), Processing local seismological waveform datasets for moment tensor inversion using mtuq. Zenodo. <https://doi.org/10.5281/zenodo.10499910>
2. Mahanta, R. and **V. Silwal** (2024). Moment tensor inversion and uncertainty analysis for 40 Uttarakhand Earthquakes (2010-2022). Zenodo. <https://doi.org/10.5281/zenodo.12541669>
3. **V. Silwal**. Seismic moment tensor catalog for crustal earthquakes in the Cook Inlet and Susitna region of southern Alaska, 2018. ScholarWorks@UA at <http://hdl.handle.net/11122/8383>: descriptor file, text file of catalog, figures with waveform fits, and input weight files.
4. A. Lomax, **V. Silwal**, and C. Tape. Hypocenter estimation for 14 earthquakes in south-central Alaska (1929–1975), 2018. ScholarWorks@UA at <http://hdl.handle.net/11122/8380>: descriptor file and zipped set of text files for each earthquake.
5. **V. Silwal**. Seismic moment tensors for six events in the Minto Flats fault zone, 2012–2016, 2018. ScholarWorks@UA at <http://hdl.handle.net/11122/8253>: descriptor file, text file of catalog, figures with waveform fits, and input weight files.
6. C. Tape, **V. Silwal**, and S. Holtkamp. Step-response signals recorded during earthquakes in Alaska, 2017. ScholarWorks@UA at <http://hdl.handle.net/11122/7947> (last accessed 2017-10-27).
7. **V. Silwal**. Seismic moment tensor catalog for Minto Flats fault zone (2000–2014), 2015. ScholarWorks@UA at <http://hdl.handle.net/11122/6267> (last accessed 2016-01-22): descriptor file, text file of catalog, composite figures of waveform fits and depth searches.

8. **V. Silwal**. Seismic moment tensor catalog for southern Alaska, 2015. ScholarWorks@UA at <http://hdl.handle.net/11122/6025> (last accessed 2016-01-22): descriptor file, text file of catalog, figures with waveform fits, depth searches, and uncertainty analyses.

Conferences/ Abstracts

1. Singh, N. and **Silwal, V.**, Seismic Monitoring of the Satopanth Himalayan Glacier: investigating icequakes, signatures and environmental influences. Geodynamic Evolution of the Himalayas (GEH-2025), Kumaun University, Nainital, 6-10 Oct, 2025.
2. Manoj TM, and **Silwal, V.**, Effects on Seismic Waveforms due to Crustal Thickening in the Himalayas using Spectral Element Simulations, 18-20 June 2025, SEG Hyderabad, India
3. Maurya, S., **Silwal, V.**, Mahanta, R., and Yadav, R.: Earthquake Moment Tensor Inversion Using 3D Velocity Model in the Himalayas, EGU General Assembly 2025, Vienna, Austria, 27 Apr–2 May 2025, EGU25-6545, <https://doi.org/10.5194/egusphere-egu25-6545>, 2025.
4. Rahul, and **V. Silwal**, Effect of topography on synthetic waveforms for the Uttarakhand Himalayas, GSW Conference, BHU Varanasi, GSW78522, Mar 6-7, 2024.
5. Mahanta, R., **V. Silwal**, and M.L. Sharma. Body waves and surface waves derived Moment tensor Catalog for Garhwal-Kumaon Himalayas. AGU fall meeting, 2022. Online poster Sessions.
6. Mahanta, R., **V. Silwal**, and Rahul, Comparative Analysis of Moment Tensor Estimation Methods for Seismic Source Mechanism Characterization, GSW Conference, BHU Varanasi, GSW65089, Mar 6-7, 2024.
7. Mishra, S. and **Silwal, V.**, Web-Based Active Fault Interactive Database: An inventory of Faults and Earthquakes of the Indian Subcontinent, vol. 2022, Art. no. IN41B-01, 2022.
8. Kharita, A. and **Silwal, V.**, SPEAQ - A Python Toolbox for Assessing Station Performance After an Event, vol. 2021, Art. no. S25E-0291, 2021.
9. Modrak, R., Tape, C., Alvizuri, C. R., **Silwal, V.**, Thurin, J., and Cleveland, M., Force and moment tensor uncertainty quantification with 1D and 3D Earth models, vol. 2020, Art. no.S045-0004, 2020.
10. B. Chow, Y. Kaneko, **V. Silwal**, and C. Tape. Adjoint tomography of the Hikurangi subduction zone and New Zealand's North Island. 2018. Abstract T51I-0282 presented at 2018 Fall Meeting, AGU, Washington, D.C., 10-14 Dec.
11. Modrak, R., Silwal, V., Alvizuri, C. R., and Tape, C., Moment tensor estimation and uncertainty quantification using mtuq, instaseis, obspy and pymc, vol. 2018, Art. no. S53E-0458, 2018.
12. R. Modrak, **V. Silwal**, C. Alvuzuri, and C. Tape. Moment tensor inversion and uncertainty quantification using mtuq, instaseis, obspy and pymc. 2018. Abstract S53E-0458 presented at 2018 Fall Meeting, AGU, Washington, D.C., 10-14 Dec.
13. C. Tape, S. Holtkamp, **V. Silwal**, J. Hawthorne, Y. Kaneko, J. P. Ampuero, C. Ji, N. Ruppert, K. Smith, and M. E. West. Earthquake nucleation and fault slip complexity in the lower crust of central Alaska. 2018. Abstract T43B-03 presented at 2018 Fall Meeting, AGU, Washington, D.C., 10-14 Dec.
14. C. Tape, C. Alvizuri, **V. Silwal**, and W. Tape. Uncertainties for seismic moment tensors and applications to nuclear explosions, volcanic events, and earthquakes. 2017. Abstract S44B-01 presented at 2017 Fall Meeting, AGU, New Orleans, La., 11-15 Dec.
15. C. Alvizuri, **V. Silwal**, L. Krischer, and C. Tape. Estimation of full moment tensors, including uncertainties, for earthquakes, volcanic events, and nuclear explosions. 2017. Abstract T2.3-P8 presented at the CTBT Science and Technology Conference 2017, Vienna, Austria.

16. **V. Silwal**, C. Tape, and E. Casarotti. Wavefield simulations of earthquakes in Alaska for tomographic inversion. 2017. Abstract S51D-0619 presented at 2017 Fall Meeting, AGU, New Orleans, La., 11-15 Dec.
17. **V. Silwal** and C. Tape. Crustal earthquakes in the Cook Inlet and Susitna regions, southern Alaska. 2017. Abstract presented at 2017 EarthScope National Meeting, Anchorage, Alaska, May 16-18.
18. J. C. Purba, **V. Silwal**, and C. Tape. A 10-Year Catalog (2007–2016) of Seismic Moment Tensors in Southern Alaska. 2017. Abstract presented at 2017 EarthScope National Meeting, Anchorage, Alaska, May 16-18.
19. C. Tape, S., Holtkamp, **V. Silwal**, J. Hawthorne, Y. Kaneko, J. P. Ampuero, N. Ruppert, K. Smith, and M. E. West. Slow-to-fast earthquake nucleation in the lower crust of central Alaska. 2017. Abstract presented at 2017 EarthScope National Meeting, Anchorage, Alaska, May 16-18.
20. C. Tape, **V. Silwal**, and S. Holtkamp. An unwanted long-period step response: examples from Alaska earthquakes. 2017. Abstract presented at 2017 EarthScope National Meeting, Anchorage, Alaska, May 16-18.
21. C. Tape, A. Lomax, **V. Silwal**, J. D. Agnew, and B. Brettschneider. The Ms7.3 earthquake of August 27, 1904 in central Alaska. 2017. Abstract presented at 2017 EarthScope National Meeting, Anchorage, Alaska, May 16-18.
22. C. Alvizuri, **V. Silwal**, L. Krischer, and C. Tape. Estimation of full moment tensors, including uncertainties, for earthquakes, volcanic events, and nuclear tests. 2016. Abstract S31A-2701 presented at 2016 Fall Meeting, AGU, San Francisco, Calif., 12-16 Dec.
23. **V. Silwal** and C. Tape. Wavefield simulations of earthquakes in southern Alaska for tomographic inversion. 2016. Poster presented at 2016 CIG All Hands Meeting, Davis, Calif., 20-22 June.
24. **V. Silwal** and C. Tape. Wavefield simulations of earthquakes in southern Alaska for tomographic inversion. 2016. Abstract presented at 2016 IRIS Workshop, Vancouver, Wash., 8-10 June.

Awards

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| 1. Brian R. Zelanka Scholarship for Best Graduate Student | 2018 |
| 2. Earthscope National Meeting Scholarship | 2017 |
| 3. Best student poster at Alaska Geological Society Conference | 2016 |
| 4. Geophysical Society of Alaska scholarship | 2015 |
| 5. SEG scholarship | 2014 |
| 6. Geophysical Institute travel grant for AGU Fall Meeting | 2013 |
| 7. INSPIRE scholarship by the Department of Science & Technology, India | 2007-2012 |

Other Activities

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| 1. (Convener) 2-day workshop on Computational Seismology | 2025 |
| 2. (Invited talk) Advanced Training on Glaciers and Glacial Lakes Observations, NIH | 2025 |
| 3. (Invited talk) Journey of seismic waves from earthquake to receivers through 3D Earth structure, NCPOR | 2024 |
| 4. (Co-convener) Seismic Hazard Brainstorming Session and Workshop, ICED, IITR | 2024 |
| 5. TOTAL Well Log Analysis Course | 2011 |
| 6. Schlumberger PETREL Seismic Visualization and Interpretation Course | 2010 |